

Voice Signal Analysis Using Machine Learning for Early Detection of Parkinson's Disease

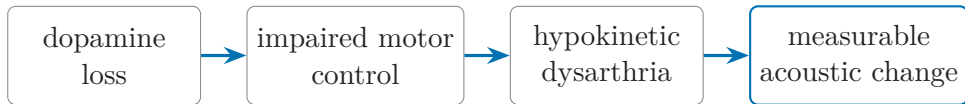
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A movement disease reaches the voice



every muscle — breathing, larynx, articulation

The weakness that shakes a hand also moves the voice.

The voice is a cheap, early window

~90%

of patients show
speech changes

5 yr

before diagnosis,
in one documented case

\$

a microphone:
non-invasive, repeatable

Not a diagnosis — a reason to see a doctor earlier.

Can a computer measure a voice recording
and tell whether its pattern resembles
Parkinson's patients — or healthy speakers?

Six stages, one shared implementation



The same code runs in training, testing, and prediction — enforced, not promised.

Every measurement points at physiology



monotone speech —————> pitch variation

unstable vocal-fold control —————> jitter, shimmer

incomplete glottal closure —————> HNR, CPPS

disturbed timing —————> pause statistics

imprecise articulation —————> MFCC dynamics

All 74 measurements implemented in this project — none downloaded.

We chose *not* to clean the sound

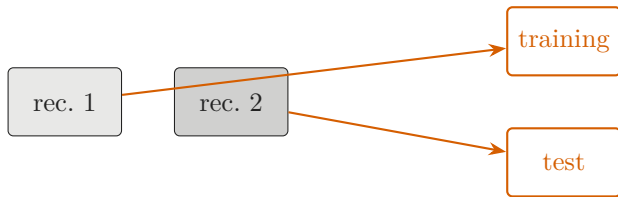


- Denoising suppresses irregularity.
- Irregularity *is* the evidence.

Cleaning would make a disordered voice look artificially healthy.

A student with the exam questions proves nothing

one person, two recordings



the model can win by
recognising the *person*

This is **data leakage**. Our program halts if it ever happens.

Validation design decides the number

0.909

balanced accuracy

an impressive result?

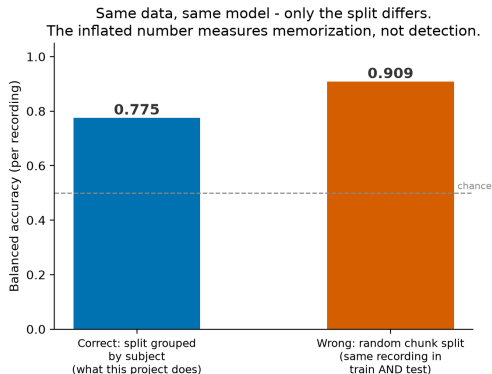
Validation design decides the number

0.909

deliberately wrong split
demonstration only

0.775

correct split, same model



The difference is memorisation, not detection — and it is why 95%+ results exist.

The rules were fixed before the experiments

- identical splits for every variant
- adoption margin: +0.02
- any result ≥ 0.95 : stop, investigate
- all 19 configurations reported

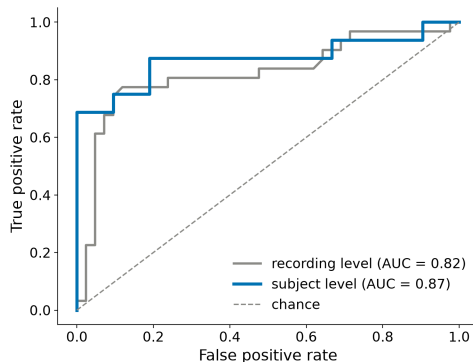
Best score 0.840 — rejected. It won by 0.018: less than the noise.

Judged on people it has never heard



- balanced accuracy **0.822 ± 0.032**
- sensitivity 0.771 specificity 0.873
- AUC **0.864**

“always healthy” scores 0.575 accuracy,
finds nobody — balanced accuracy 0.500.



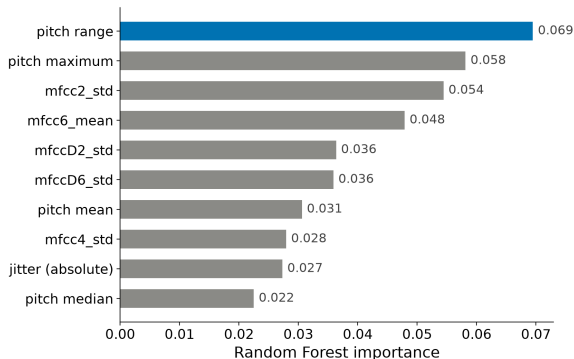
It rediscovered a known clinical sign



Top measurement:

pitch range

the acoustic correlate of
monotone speech —
described by clinicians
for decades.



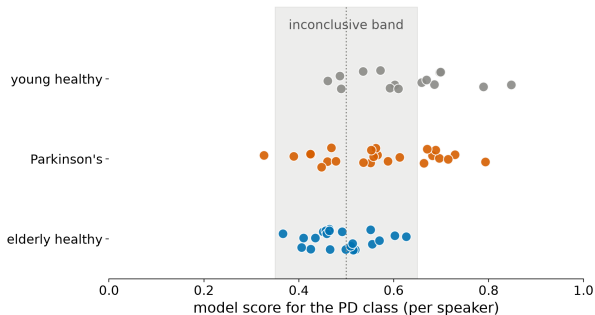
Italy: a frozen model, a trap, and a puzzle



Trap caught: every 44.1 kHz file was a patient. Band-limited all audio to 16 kHz.

Result: AUC **0.701**, balanced accuracy 0.629.

Puzzle: healthy young 0.626 > patients 0.566. Why?



A system that refuses to guess

Parkinson's-Related Voice Changes — Research Prototype

This is a research screening-support prototype and is not a medical diagnostic tool. Its result must not replace evaluation by a qualified healthcare professional.

Select a voice recording (WAV format). The system extracts acoustic features (F0, jitter, shimmer, HNR, MFCC) with the same pipeline used in training, and applies a machine learning model trained on the MDVR-KCL research dataset.

> What do HC and PD mean here?

How do you want to provide the recording?

☒ Upload a WAV file ☐ Record with the microphone

1007_pd_2_0_0.wav
35.6 MB

File information

- File: 1007_pd_2_0_0.wav
- Duration: 147.7 s (analyzed after silence trimming: 145.3 s)
- Original sample rate: 44100 Hz

Research model result

The acoustic pattern was classified by the research model as closer to the PD class.

Model score for the PD class: 0.96 (0 = closer to HC class, 1 = closer to PD class; scores between 0.35 and 0.65 are reported as inconclusive). This is a property of the research model, not a person's medical risk or probability of disease.

This is a research screening-support prototype and is not a medical diagnostic tool. Its result must not replace evaluation by a qualified healthcare professional.

- scores 0.35–0.65 $\Rightarrow$ “inconclusive”
- 100% of elderly healthy Italian speakers landed there
- declining to answer is a feature

Optional: live demonstration

- safest demo: a corpus recording, command line
- analysis takes about one minute
- fallback: the screenshots on the previous slide carry the same content

skip this slide if time is short

What this is — and is not

- not a diagnostic tool; many conditions change a voice
- 37 subjects, one language, one device

**Validation design moves the number
more than model engineering does.**

An honest **0.822** is worth more than an inflated one.

Thank you

Questions welcome.

Backup: why 82% and not 95%?

- published 95%+ figures usually split recordings or windows, not people
- our own pipeline: 0.909 leaky vs 0.775 honest — same model
- 0.822 is measured on people the model never heard

Backup: why not deep learning?

- 37 subjects is far below what deep networks need
- examiners can ask this model *which* measurements it used
- our small neural network collapsed in the confounder test ($0.671 \rightarrow 0.526$)

Backup: why balanced accuracy?

- classes are imbalanced: 42 HC vs 31 PD recordings
- “always healthy”: 0.575 accuracy, zero patients found
- balanced accuracy gives that model 0.500 — exposed

Backup: why reject the 0.840 ensemble?

- adoption margin +0.02 was fixed before any experiment
- it won by 0.018; seed-to-seed noise is about ± 0.03
- picking the most complex of tied variants fits the validation, not the problem

Backup: can it diagnose Parkinson's?

- no — and it is not designed to
- colds, ageing, smoking, fatigue also change the voice
- it reports similarity to dataset groups; clinical evaluation stays with clinicians

Backup: is 37 subjects too few?

- yes — it is the dominant limitation, stated openly
- mitigated: grouped validation, 3 seeds, spread reported (± 0.032)
- plus an external test on 61 further speakers

Backup: is it detecting male vs female voices?

- absolute pitch encodes speaker sex — so we removed those four features
- selected model held: $0.768 \rightarrow 0.780$
- the MLP collapsed ($0.671 \rightarrow 0.526$) and was excluded

Backup: hardest technical problem?

- finding confounders that would flatter the results
- Italian corpus: every 44.1 kHz file was a patient — bandwidth, not disease
- caught at inspection; all audio band-limited to 16 kHz first